

MH2100 Calculus III

Midterm Exam 1 (Solutions)

AY 2024/2025, Semester 1

Quantitative Research Society @NTU

September 19, 2024

Overview

- **Coverage:** curves and tangent vectors; limits and differentiability; linear approximation; the Multivariable Chain Rule; tangent planes; implicit differentiation; and extrema.
- **Historical source:** MH2100 Midterm Exam 1, AY2024/2025 Semester 1. Every problem below is transcribed from the official paper. Solutions and marking schemes are unofficial QRS@NTU material.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20 (110 minutes), venue TBA. One double-sided typed A4 sheet is permitted; attachments and sticky notes are not.

Instructions to candidates (historical, verbatim)

- (1) DO NOT TURN OVER THE EXAM PAPER UNTIL INSTRUCTED TO DO SO.
- (2) This midterm exam consists of ONLY THREE problems, each consisting of multiple parts.
- (3) Candidates are required to attempt ALL THREE problems.
- (4) The total number of points for each problem is indicated at the top right-hand corner of its page.
- (5) Candidates may write anywhere on the blank pages reserved for showing work.
- (6) A SINGLE double-sided A4-sized help-sheet with handwritten notes is permitted.

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Question 1**(15 Points)****Problem 1**

- (a) A curve
- C
- in
- $\mathbb{R}^2$
- is parametrized by the vector function
- $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$
- , defined by

$$\mathbf{r}(t) = (t^2 - 1, t^3 - t).$$

[4 points] Find all the unit tangent vectors to C produced by $\mathbf{r}$ at the point $(0, 0)$ on C .

- (b) For each of the following limits, either evaluate it or explain why it does not exist.

- (i) [2 points]

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}.$$

- (ii) [2 points]

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6}.$$

- (iii) [2 points]

$$\lim_{(x,y) \rightarrow (1,2)} \frac{\ln(x)}{4x^2 - y^2}.$$

- (c) Define a function
- $f : \mathbb{R}^2 \rightarrow \mathbb{R}$
- by

$$f(x, y) = x^2 + \sin(xy).$$

[5 points] Find all the unit vectors $\mathbf{u}$ in $\mathbb{R}^2$ such that $(D_{\mathbf{u}}(f))(1, 0) = 1$.**Solution****Method 1: Direct calculation**

- (a)
- $\mathbf{r}(t) = (0, 0) \iff t = \pm 1$
- :

$$\mathbf{r}'(t) = \langle 2t, 3t^2 - 1 \rangle \implies \mathbf{r}'(1) = \langle 2, 2 \rangle, \mathbf{r}'(-1) = \langle -2, 2 \rangle$$

$$\implies \boxed{\frac{1}{\sqrt{2}} \langle 1, 1 \rangle \quad \text{and} \quad \frac{1}{\sqrt{2}} \langle -1, 1 \rangle}$$

- (b) (i)
- $\left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y| \rightarrow 0$
- (Squeeze)
- $\implies \boxed{0}$

- (ii)

$$x = 0 : 0, \quad x = y^3 : \frac{y^6}{2y^6} = \frac{1}{2} \implies \boxed{\text{the limit does not exist}}$$

- (iii)

$$x = 1 : 0, \quad y = 2 : \lim_{x \rightarrow 1} \frac{\ln x}{4x^2 - 4} = \lim_{x \rightarrow 1} \frac{1/x}{8x} = \frac{1}{8} \\ \implies \boxed{\text{the limit does not exist}}$$

- (c)
- $\nabla f = \langle 2x + y \cos(xy), x \cos(xy) \rangle$
- ,
- $\nabla f(1, 0) = \langle 2, 1 \rangle$
- :

$$\mathbf{u} = \langle a, b \rangle : \quad 2a + b = 1, \quad a^2 + b^2 = 1$$

$$b = 1 - 2a \implies a(5a - 4) = 0$$

$$(D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u})$$

$$\implies \boxed{\mathbf{u} = \langle 0, 1 \rangle \quad \text{or} \quad \mathbf{u} = \left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle}$$

UNOFFICIAL QRS marking scheme (15 points)

- (a) **(4 points)** 1: finds both parameters; 1: differentiates; 1: evaluates both velocities; 1: gives the two oriented unit tangents.
- (b) **(6 points)**
 - (i) **(2 points)** 1: valid bound; 1: limit 0.
 - (ii) **(2 points)** 1: two paths; 1: nonexistence conclusion.
 - (iii) **(2 points)** 1: two valid path limits; 1: nonexistence conclusion.
- (c) **(5 points)** 1: gradient; 1: evaluates it; 1: directional-derivative condition; 1: unit condition; 1: both vectors.

Question 2**(15 Points)****Problem 2**(a) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = \begin{cases} \frac{2x^2y + y^3}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

(i) [3 points] Determine whether f is continuous at $(0, 0)$. Justify your answer.(ii) [2 points] Evaluate $f_x(0, 0)$ and $f_y(0, 0)$, if these quantities exist.(iii) [4 points] Determine whether f is totally differentiable at $(0, 0)$. Justify your answer.(b) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = e^{2x^2+3y}.$$

[3 points] Use a linear approximation of f at $(0, 0)$ to estimate the value of $f(0.01, -0.02)$.(c) Suppose that the functions $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ satisfy the following properties:

- f is totally differentiable throughout $\mathbb{R}^3$.
- For all $x, y, z \in \mathbb{R}$, we have $g(x, y, z) = f(x - y, y - z, z - x)$.

[3 points] Use the Multivariable Chain Rule to show, for all $x, y, z \in \mathbb{R}$, that

$$g_x(x, y, z) + g_y(x, y, z) + g_z(x, y, z) = 0.$$

Solution**Method 1: Direct calculation**(a) (i) $r = \sqrt{x^2 + y^2}$:

$$|f(x, y)| \leq \frac{2x^2|y| + |y|^3}{r^2} \leq 3r \rightarrow 0 = f(0, 0) \quad (\text{Squeeze})$$

$$\implies \boxed{f \text{ is continuous at } (0, 0).}$$

(ii) $f(h, 0) = 0, f(0, h) = h$ ($h \neq 0$) $\implies f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0)}{h} = 0, f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h)}{h} = 1$
 1: $\boxed{f_x(0, 0) = 0, f_y(0, 0) = 1}$

(iii) Suppose f totally differentiable at $(0, 0) \implies L(x, y) = f_x(0, 0)x + f_y(0, 0)y = y$.

$$f(t, t) - L(t, t) = \frac{3t^3}{2t^2} - t = \frac{t}{2}$$

$$\frac{f(t, t) - L(t, t)}{\sqrt{t^2 + t^2}} = \frac{1}{2\sqrt{2}} \not\rightarrow 0 \quad (t \rightarrow 0^+)$$

$$\implies \boxed{f \text{ is not totally differentiable at } (0, 0).}$$

(b) $f_x = 4xe^{2x^2+3y}, f_y = 3e^{2x^2+3y}$; at $(0, 0)$: $f = 1, f_x = 0, f_y = 3$:

$$L(x, y) = 1 + 3y \implies \boxed{f(0.01, -0.02) \approx L(0.01, -0.02) = 0.94}$$

(c) Chain Rule (f totally differentiable), $f_i := \partial_i f(x - y, y - z, z - x)$:

$$g_x = f_1 - f_3, \quad g_y = -f_1 + f_2, \quad g_z = -f_2 + f_3$$

$$\implies \boxed{g_x(x, y, z) + g_y(x, y, z) + g_z(x, y, z) = 0}$$

Method 2: translation invariance

(c) $g = f \circ (\text{linear map})$ is totally differentiable:

$$g(x + s, y + s, z + s) = f(x - y, y - z, z - x) = g(x, y, z) \quad \forall s \in \mathbb{R}$$
$$0 = \frac{d}{ds} \Big|_{s=0} g(x + s, y + s, z + s) = g_x + g_y + g_z \quad (\text{Chain Rule, } g \text{ totally differentiable})$$

UNOFFICIAL QRS marking scheme (15 points)

(a) **(9 points)**

(i) **(3 points)** 2: valid bound; 1: continuity.

(ii) **(2 points)** 1: each partial derivative.

(iii) **(4 points)** 1: candidate derivative; 2: diagonal remainder; 1: non-differentiability conclusion.

(b) **(3 points)** 1: derivatives at the base point; 1: linear approximation; 1: estimate 0.94.

(c) **(3 points)** 2: three Chain Rule derivatives; 1: sums them to zero.

Question 3

(20 Points)

Problem 3

- (a) [4 points] Find a linear equation of the tangent plane to the surface S in $\mathbb{R}^3$, given by

$$xy^2z^3 + x^4y^3z^2 = 4,$$

at the point $(1, -1, 2)$.

- (b) Define a function $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$F(x, y, z) = x^3 + z^2 + ye^{xz} + z \cos(y).$$

Let $S = \{(x, y, z) \in \mathbb{R}^3 \mid F(x, y, z) = 0\}$, which is the level surface of F with value 0.

- (i) [2 points] Show that the quantity $F_z(0, 0, 0)$ exists and is non-zero. **Remark (safe to ignore):** By the Implicit Function Theorem, we can thus find

- a sufficiently small open disk D in $\mathbb{R}^2$ centered at $(0, 0)$,
- a sufficiently small bounded open interval I in $\mathbb{R}$ centered at 0, and
- a function $z : D \rightarrow I$ that is totally differentiable throughout D ,

such that, within the small region $D \times I$, the surface S is actually the graph surface of z .

- (ii) [4 points] Use the method of implicit differentiation to evaluate $\frac{\partial z}{\partial x}(0, 0)$ and $\frac{\partial z}{\partial y}(0, 0)$.

- (c) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = 12x^2 + y^3 - 12xy.$$

- (i) [3 points] Find all the critical points of f .
- (ii) [4 points] Use the Second-Derivative Test to determine the nature of each critical point of f .
- (iii) [3 points] Determine whether f has any absolute extremum. Justify your answer.

Solution

Method 1: Direct calculation

- (a) $G := xy^2z^3 + x^4y^3z^2$, $S = \{G = 4\}$, normal ∇G :

$$\nabla G = \langle y^2z^3 + 4x^3y^3z^2, 2xyz^3 + 3x^4y^2z^2, 3xy^2z^2 + 2x^4y^3z \rangle$$

$$\nabla G(1, -1, 2) = \langle -8, -4, 8 \rangle$$

$$-8(x - 1) - 4(y + 1) + 8(z - 2) = 0$$

$$\implies \boxed{-2x - y + 2z = 3}$$

- (b) (i)

$$F_z = 2z + xye^{xz} + \cos y, \quad F_z(0, 0, 0) = 1 \neq 0$$

$$\implies \boxed{F_z(0, 0, 0) \text{ exists and is non-zero.}}$$

(ii) Implicit differentiation, $z_x = -F_x/F_z$, $z_y = -F_y/F_z$ ($F_z(0,0,0) \neq 0$):

$$\begin{aligned} F_x &= 3x^2 + yze^{xz}, & F_y &= e^{xz} - z \sin y \\ (F_x, F_y, F_z)(0,0,0) &= (0,1,1) \\ \implies & \boxed{\frac{\partial z}{\partial x}(0,0) = 0, \quad \frac{\partial z}{\partial y}(0,0) = -1} \end{aligned}$$

(c) (i)

$$\begin{aligned} f_x &= 24x - 12y = 0 \implies y = 2x \\ f_y &= 3y^2 - 12x = 0 \implies y^2 = 4x, \quad y = 2x \implies x(x-1) = 0 \\ \implies & \boxed{(0,0) \quad \text{and} \quad (1,2)} \end{aligned}$$

(ii) $f_{xx} = 24$, $f_{xy} = -12$, $f_{yy} = 6y$:

$$\begin{aligned} D &= f_{xx}f_{yy} - f_{xy}^2 = 144(y-1) \quad (\text{Second Derivative Test}) \\ (0,0) : D &< 0 \implies \text{saddle} \\ (1,2) : D &> 0, \quad f_{xx} > 0 \implies \text{local minimum} \\ \implies & \boxed{(0,0) \text{ is a saddle point; } (1,2) \text{ is a local minimum}} \end{aligned}$$

(iii)

$$\begin{aligned} f(0,y) &= y^3 \text{ unbounded above and below} \\ \implies & \boxed{f \text{ has no absolute maximum and no absolute minimum}} \end{aligned}$$

UNOFFICIAL QRS marking scheme (20 points)

(a) **(4 points)** 2: gradient; 1: evaluates normal; 1: correct plane.

(b) **(6 points)**

(i) **(2 points)** 1: computes F_z ; 1: evaluates it as nonzero.

(ii) **(4 points)** 2: computes F_x, F_y ; 1: implicit-derivative formulae; 1: both values.

(c) **(10 points)**

(i) **(3 points)** 2: solves first-derivative equations; 1: both points.

(ii) **(4 points)** 1: second derivatives; 1: discriminant; 1: saddle; 1: local minimum.

(iii) **(3 points)** 2: unbounded path; 1: no absolute extrema.